SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
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B.Tech in Computer Science and Engineering

Course: Business Intelligence
Assessment: CA3 — Case Study Using R Programming

Renewable Energy Intelligence: Solar Generation, Electricity Demand and Energy Performance Analytics Using R

A Business Intelligence Case Study Based on the India Grid Analytics R Shiny Dashboard

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1. Abstract

Growing shares of solar and wind generation make it harder for energy planners to see when renewable output covers electricity demand and when other resources must fill the remainder. This case study documents Renewable Energy Intelligence, an interactive R Shiny dashboard that analyses India's national electricity demand together with solar and wind generation at one-hour resolution. The underlying data are the Mendeley Data collection of Mukherjee et al. (2025), compiled from Grid-India (NERLDC) records; the application covers January 2024 to June 2025. Built in R with Shiny, plotly and DT, the dashboard offers seven views: a KPI overview, generation analysis, generation versus demand (residual demand), variability indicators, a daily demand forecast with holdout validation, business insights, and a filterable data table with CSV export. The screenshots supplied for this report show a filtered window of weekend days from 25 May to 24 June 2025. In that window, solar and wind together supplied 17.65% of demand, solar output peaked at 61.71 GW and was near zero overnight, and hourly residual demand rose by as much as 29.11 GW in a single hour. The demand forecast reported a holdout MAPE of 6.16%. The findings indicate where flexibility and storage evaluation should concentrate, but they rest on a narrow window and do not measure grid shortages or solar panel losses.

2. Introduction and Business Problem

Business Intelligence (BI) converts operational time series into indicators, comparisons and interactive views that support planning decisions. In power systems, operators must balance generation and demand continuously, while planners decide where to add flexibility, storage or forecasting capability. Solar output follows daylight and is absent at night, wind output follows weather and season, and electricity demand follows daily and seasonal routines. The quantity that remains after subtracting solar and wind from demand, known as residual or net demand, is what dispatchable thermal, hydro, nuclear, storage and inter-regional transfers must supply. How quickly that quantity changes from one hour to the next determines how much ramping flexibility the rest of the system needs.

The business problem addressed here is the absence of a single, interactive view that shows, for a chosen period, season and day type, (i) how much hourly national demand is covered by solar and wind, (ii) when solar output disappears, (iii) how fast residual demand changes, and (iv) how demand may evolve over the next days. The intended users are renewable energy analysts, energy planners and grid operations teams who need descriptive evidence to prioritise further study. The application is an exploratory analytics tool; it is not a dispatch or market-operation system.

3. Objectives and Scope

The implemented project pursues the following objectives:

1. Summarise solar, wind and combined renewable generation against electricity demand using KPI indicators (Overview tab).
2. Characterise the diurnal and seasonal or monthly behaviour of solar and wind generation (Generation Analysis tab).
3. Quantify residual demand, one-hour ramp rates and the distribution of renewable penetration (Generation vs Demand tab).
4. Calculate variability and correlation indicators for solar, wind and demand (Energy Performance tab).
5. Forecast daily electricity demand with a user-selected horizon and holdout validation (Forecasting tab).
6. Present findings, indicative recommendations and exportable filtered data through an interactive Shiny dashboard (Business Insights and Data Explorer tabs).

Scope. Geographically, the data represent India as a whole, with the five regional grids combined (Mukherjee et al., 2025). Temporally, the dashboard footer states coverage of January 2024 to June 2025 at one-hour intervals. All screenshots supplied for this report share one filter state (25 May to 24 June 2025, all four seasons selected, Weekends only), so every number reported below describes that subset. The analysis is limited to national aggregates of demand, solar and wind; it contains no plant-level, weather, installed-capacity, storage or other-generation data.

4. Dataset Description and Preparation

The dataset is *Electricity Demand, Solar and Wind Generation Data (September 2021–June 2025) of India at 1-hour interval* (Version 2), published on Mendeley Data on 15 September 2025 by Debanjan Mukherjee, Karuna Kalita and Subhash Kumar (Mukherjee et al., 2025). The repository describes it as phasor-measurement-unit (PMU) based electricity demand, solar and wind generation for the combined five regional grids of India, provided

by the North-Eastern Regional Load Despatch Centre (NERLDC), Grid-India, under the Ministry of Power. Table 1 summarises the dataset as used in this project. The repository title spans September 2021 to June 2025, whereas its description text mentions September 2021 to December 2023; this report follows the coverage stated inside the application (January 2024 to June 2025).

Table 1. Dataset summary

Item	Detail
Source	Mukherjee, Kalita and Kumar (2025), Mendeley Data, Version 2, https://doi.org/10.17632/y58jknpgs8.2
Original data provider	NERLDC, Grid Controller of India Limited (Grid-India), Ministry of Power, Government of India
Geographic coverage	India, five regional grids combined (national aggregate)
Period used in the application	January 2024 to June 2025 (application footer); screenshots analysed: 25 May to 24 June 2025, weekends only
Granularity	Hourly (1-hour interval); hour of day expressed in IST in the charts
Records	Full application dataset:. Filtered window in screenshots: 216 hourly records (derived: 9 weekend days × 24 h; see Section 7.1)

Note: the record count of the filtered window is derived by the author from the calendar and the KPI values; it is not displayed by the application.

Table 2 lists the analysis-ready fields visible in the Data Explorer tab (Figure 1). The application shows demand, solar and wind in gigawatts (GW), and its metadata panel defines the raw series as Demand_MW, Solar_MW and Wind_MW in megawatts; the conversion therefore appears to be a division by 1,000 (inference). The derived fields were checked against the displayed rows: for the first record, 17.60 GW ÷ 191.92 GW = 9.17% and 191.92 − 17.60 = 174.32 GW; for the second, 165.20 − 174.32 = −9.12 GW. All three agree with the table. The first row also shows a ramp of −8.28 GW although no preceding record is displayed, which indicates that ramp values are calculated on the full chronological series before filters are applied (inferred).

Table 2. Analysis-ready fields visible in the Data Explorer table

Field	Unit	Definition (as displayed in the application metadata)
TIMESTAMP	date-time	Hourly timestamp
DEMAND_GW	GW	National electricity demand
SOLAR_GW / WIND_GW	GW	Solar generation / wind generation
RENEWABLE_GW	GW	Solar + wind generation
RE_SHARE	%	(Renewable generation ÷ demand) × 100
NET_DEMAND_GW	GW	Residual demand: demand − renewable generation
NET_RAMP_GW	GW/h	Residual demand at hour t minus residual demand at hour t − 1
SEASON	category	Season label (column truncated in the screenshot; further columns, if any, are not visible)

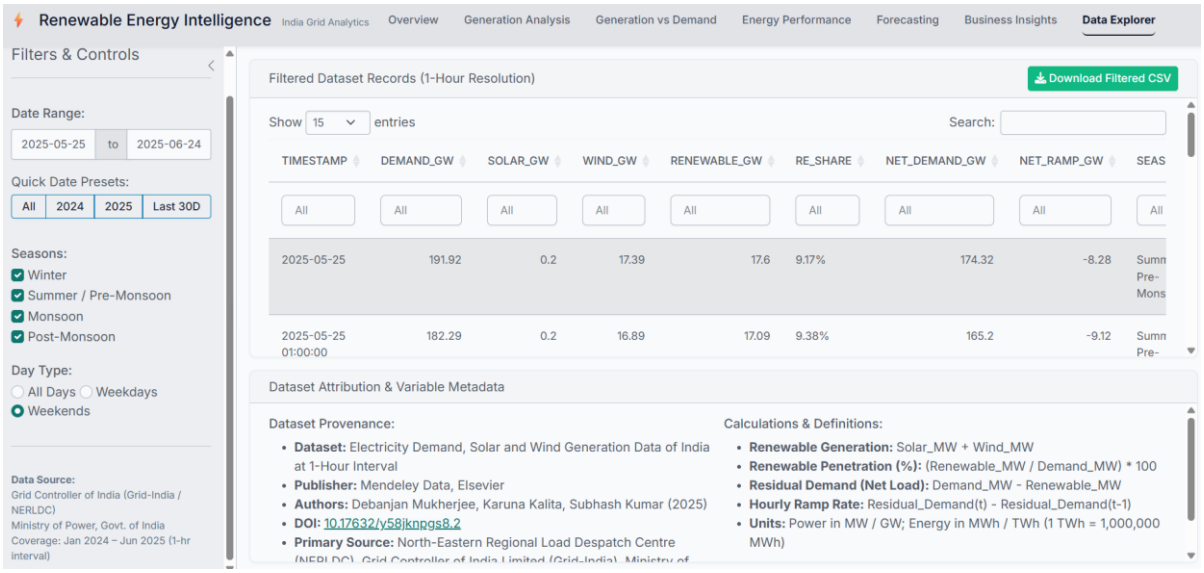


Figure 1. Data Explorer tab showing the filtered hourly records, the CSV download button and the dataset attribution and variable metadata panel.

Source: Author's own screenshot of the Renewable Energy Intelligence application, developed using R Shiny.

Data preparation in R therefore consists of loading the hourly records, parsing timestamps, converting units, assigning season and day-type labels for the filters, and deriving renewable generation, penetration, residual demand and ramp (Section 5). The cleaning steps themselves are [confirm from the R script: handling of missing values, duplicates and outliers].

5. Methodology and R Programming Implementation

Data acquisition → Data cleaning → Data transformation → KPI calculation → Exploratory analysis → Visualisation → Business interpretation

The application is written in R (R Core Team, n.d.). The interface provides direct evidence for the following packages. **Shiny** (Posit Software, PBC, n.d.) supplies the reactive framework in which the sidebar inputs (date range, preset buttons, season checkboxes, day-type radio buttons, sliders) drive every output. The navigation bar, collapsible sidebar, cards and value boxes are consistent with the **bslib** package (Sievert, Cheng, et al., n.d.). The toolbar visible on the charts in Figures 3 to 6 (zoom, pan, camera) is the signature of **plotly** (Sievert, Parmer, et al., n.d.), and the tables with entry selector, search box, column filters, sortable headers and pagination are **DT** tables (Xie et al., n.d.). Data-manipulation, date-time and forecasting packages are [confirm from the R script, e.g. dplyr, lubridate and the forecasting package] and are not asserted here.

The indicators below follow the definitions displayed in the application (Figures 1 and 4). Hourly renewable generation, penetration, residual demand and ramp are:

$$RE_t = Solar_t + Wind_t \quad (1)$$

$$Penetration_t\,(\%) = (RE_t \div Demand_t) \times 100 \quad (2)$$

$$Residual_t = Demand_t - RE_t \quad (3)$$

$$Ramp_t = Residual_t - Residual_{t-1} \quad (4)$$

Equation 3 is the generation–demand gap. It is a renewable shortfall relative to demand, not an electricity supply deficit: the application states explicitly that a net demand gap does not imply a shortage or blackout, because dispatchable thermal, hydro, nuclear, storage and inter-regional transfers balance it (Figure 4). The application does not truncate Equation 3 at zero; since the maximum renewable share in the analysed window is 36.98%, the gap is positive in every hour and a truncated version would give identical values here. Energy totals in TWh are sums of hourly GW values multiplied by one hour. As a consistency check, 216 hours × 202.57 GW = 43.76 TWh, matching the displayed 43.75 TWh of demand. Variability is reported as the coefficient of variation (CV, standard deviation divided by mean), and the solar–wind relationship as Pearson's correlation coefficient r.

The Forecasting tab accepts a target series (Electricity Demand is selected), a horizon of 7 to 30 days (14 selected) and a holdout split of 0.10 to 0.25 (0.15 selected), and reports holdout MAPE (mean absolute percentage error), RMSE (root mean square error), MAE (mean absolute error) and the coverage of the 95% prediction interval. The forecasting model, and whether the holdout is the chronologically latest 15% of the daily series, are [confirm from

the R script]; a chronological split is the appropriate design for time series, but it cannot be verified from screenshots.

6. Dashboard Design and Functionality

The application, titled *Renewable Energy Intelligence – India Grid Analytics*, is a single-page Shiny dashboard with a top navigation bar of seven tabs: Overview, Generation Analysis, Generation vs Demand, Energy Performance, Forecasting, Business Insights and Data Explorer. A collapsible left sidebar, *Filters & Controls*, persists across tabs and contains a date range, quick presets (All, 2024, 2025, Last 30D), four season checkboxes (Winter, Summer / Pre-Monsoon, Monsoon, Post-Monsoon), a day-type selector (All Days, Weekdays, Weekends) and a data-source note naming Grid-India (NERLDC) and the Ministry of Power with coverage January 2024 to June 2025. In every supplied screenshot the date range is 2025-05-25 to 2025-06-24 and Weekends is selected.

As shown in Figure 2, the Overview tab establishes the visual hierarchy: eight KPI value boxes sit at the top (total clean energy as TWh and as a share of gross demand; solar energy and peak GW; wind energy and peak GW; total demand and peak GW; average demand; maximum instantaneous renewable share; maximum one-hour net ramp; and the number of hours with renewable share below 5%). Below them are a demand-and-renewable timeline with Daily, Weekly and Hourly radio buttons and an *Executive Summary & Current Scope* panel; these two panels are only partly visible in the screenshot and their content is not described further. Detailed analyses sit on the other tabs inside titled cards, using a consistent colour coding of amber for solar and blue for wind (Figure 3). The Data Explorer tab (Figure 1) adds a DT table with entry selector, search, per-column filters and a *Download Filtered CSV* button, plus a provenance and definitions panel that supports transparency. Together, the shared filters and linked views let a user move from headline indicators to distributions and underlying records for the same period without leaving the application.

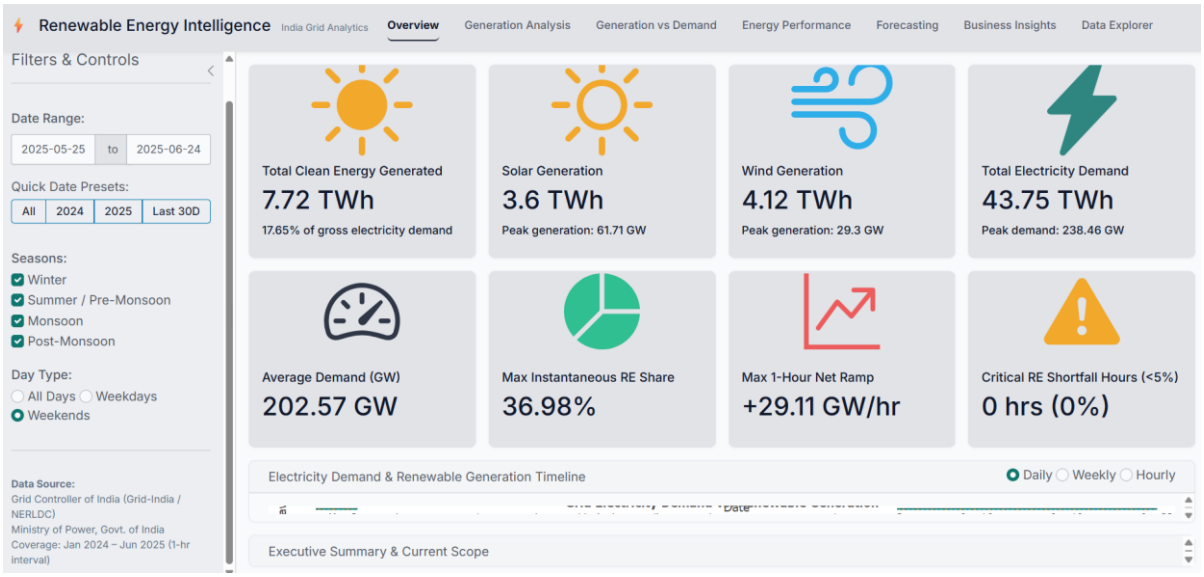


Figure 2. Overview tab with the Filters & Controls sidebar and eight KPI value boxes for 25 May–24 June 2025 (weekends).

Source: Author's own screenshot of the Renewable Energy Intelligence application, developed using R Shiny.

7. Results and Discussion

All values in this section describe the filter state visible in the screenshots: 25 May to 24 June 2025, weekends only, which comprises nine days (25 and 31 May; 1, 7, 8, 14, 15, 21 and 22 June). Values marked *derived* were calculated by the author from displayed numbers.

7.1 Overall renewable contribution

Table 3 reorganises the Overview indicators shown in Figure 2. **Observed:** solar and wind together supplied 7.72 TWh, or 17.65% of the 43.75 TWh of demand; wind contributed slightly more energy than solar (4.12 against 3.60 TWh), and no hour had a renewable share below 5%. **Significance:** wind output appears to provide a floor of contribution through the night, whereas solar adds a concentrated midday contribution; more than 82% of the energy still had to come from other sources, so planning attention belongs to residual demand rather than to the renewable total. **Implication:** the maximum instantaneous share of 36.98% means renewable output never

approached total demand in this window, so there is no evidence here of midday over-generation relative to national demand. **Limitation:** "clean energy" in the dashboard means solar plus wind only and excludes hydro and nuclear, so 17.65% must not be read as the total low-carbon share; and nine weekend days cannot represent the full year.

Table 3. Overview indicators for 25 May–24 June 2025 (weekends) and derived readings

Indicator	Displayed value	Reading
Total electricity demand	43.75 TWh; peak 238.46 GW	Average 202.57 GW; peak-to-average ratio 1.18 (derived)
Total clean energy (solar + wind)	7.72 TWh (17.65% of demand)	Excludes hydro and nuclear
Solar generation	3.60 TWh; peak 61.71 GW	46.6% of renewable energy (derived)
Wind generation	4.12 TWh; peak 29.30 GW	53.4% of renewable energy (derived)
Maximum instantaneous RE share	36.98%	Renewables never exceeded 37% of hourly demand
Maximum one-hour net ramp	+29.11 GW/h	Largest hourly rise in residual demand
Hours with RE share below 5%	0 h (0%)	Renewable contribution never negligible

7.2 Diurnal and seasonal generation patterns

Figure 3 shows the diurnal profile, the seasonal totals and the peak milestones. **Observed:** mean solar output forms a bell shape centred at about 12:00–13:00 IST and is close to zero overnight, whereas mean wind output stays near 20 GW across the day within a 10th–90th percentile band. The highest solar hour was 12:00 on 8 June 2025 at 61.71 GW against demand of 211.55 GW; solar alone then covered about 29% of demand (derived), and the displayed renewable share for that hour is about 35%. **Significance and implication:** solar timing is predictable but confined to roughly ten daylight hours, so evening and night supply depends on wind and non-renewable sources. **Limitation:** the seasonal chart shows summed energy, not averages. Assuming May falls in Summer / Pre-Monsoon and June in Monsoon, the window holds 2 and 7 weekend days respectively, and the solar bars (about 0.8 and 2.8 TWh, read from the chart) are in almost exactly that proportion; the difference between the bars therefore reflects the number of days, not seasonal intensity. The duration-curve panel is compressed in this screenshot and is not interpreted quantitatively.

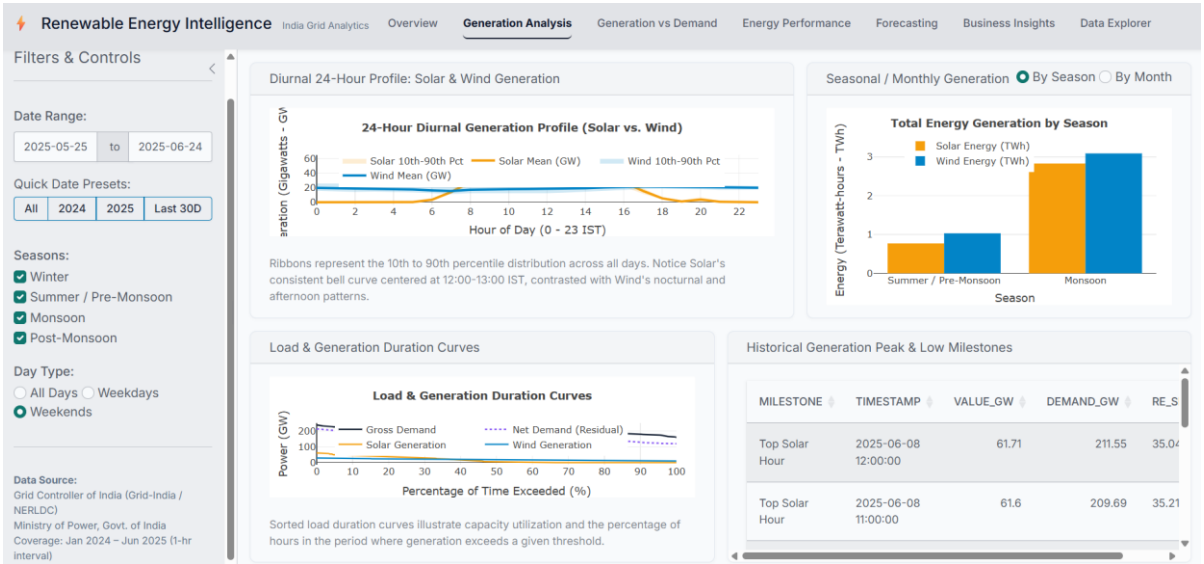


Figure 3. Generation Analysis tab: 24-hour solar and wind profile, seasonal energy totals, load and generation duration curves, and peak milestones.

Source: Author's own screenshot of the Renewable Energy Intelligence application, developed using R Shiny.

7.3 Residual demand, ramps and renewable penetration

Figure 4 begins with a banner stating that residual demand equals demand minus solar and wind and that a net demand gap does not imply a shortage, which matches the distinction required in Section 5. **Observed:** the one-hour net ramp histogram is concentrated around zero with a thin right tail reaching roughly +25 to +30 GW, consistent with the maximum of +29.11 GW/h in Table 3; the penetration histogram spans about 5% to 37%, with its tallest bar near 10%. The duck-curve panel is rendered too compressed in this screenshot to read values, so the evening-ramp interpretation relies on the dashboard's own annotations and on the ramp statistics. **Significance and implication:** the right tail shows the size of upward changes that dispatchable resources and storage must absorb within an hour, which is the quantity relevant to ramping-capability planning. **Limitation:** hourly data hide sub-hourly ramps, and the ramp histogram mixes solar-driven evening ramps with demand-driven ramps (for example morning pick-up), which the dashboard does not separate. The panel title "% of Hourly Demand Met" should be read as the renewable share of demand, not as the proportion of demand the system met.



Figure 4. Generation vs Demand tab: grid net load definition, duck curve, one-hour net load ramp distribution and renewable penetration distribution.

Source: Author's own screenshot of the Renewable Energy Intelligence application, developed using R Shiny.

7.4 Variability and solar–wind relationship

Figure 5 and Table 4 present the variability indicators. **Observed:** the diurnal-seasonal heatmap shows zero average solar output from about 19:00 to 06:00 and the highest values between about 10:00 and 15:00; June midday values look marginally higher than May, but May contains only two weekend days, so this is not seasonal evidence. The scatter plot gives $r = -0.115$ between solar and wind. Solar has by far the highest CV (54.73%), wind is intermediate (24.75%) and demand is the most stable (7.96%). **Significance:** $r = -0.115$ is a weak negative association ($r^2 \approx 0.013$, derived, i.e. about 1% of shared variance), so the dashboard's description of "natural anti-coincidence" overstates the evidence; it is a descriptive statistic and implies no causal mechanism. The high solar CV partly reflects the day–night cycle if night hours are included in the calculation, which the screenshot does not state. **Implication:** renewable forecasting should separate the predictable diurnal shape from weather-driven variability, which needs irradiance data. **Limitation:** CV and the "Average Daytime Solar Capacity Proxy" (45.09 GW) are computed from generation, not from installed capacity; no panel losses or conversion efficiency are measured here.

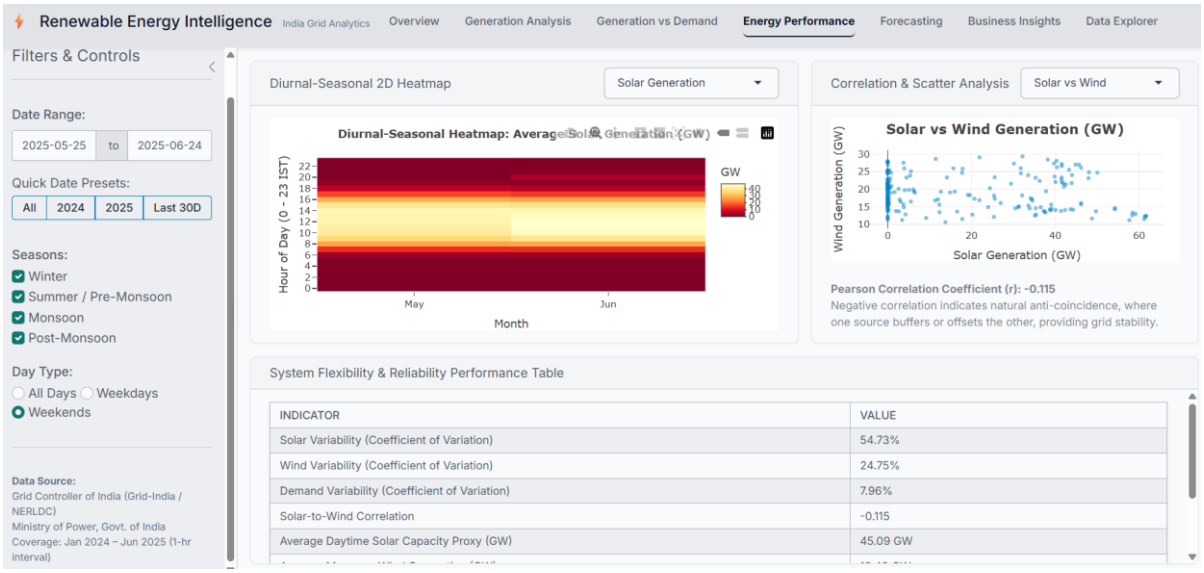


Figure 5. Energy Performance tab: diurnal-seasonal solar heatmap, solar versus wind scatter plot and system flexibility and reliability indicators.

Source: Author's own screenshot of the Renewable Energy Intelligence application, developed using R Shiny.

Table 4. Variability and correlation indicators (Energy Performance tab)

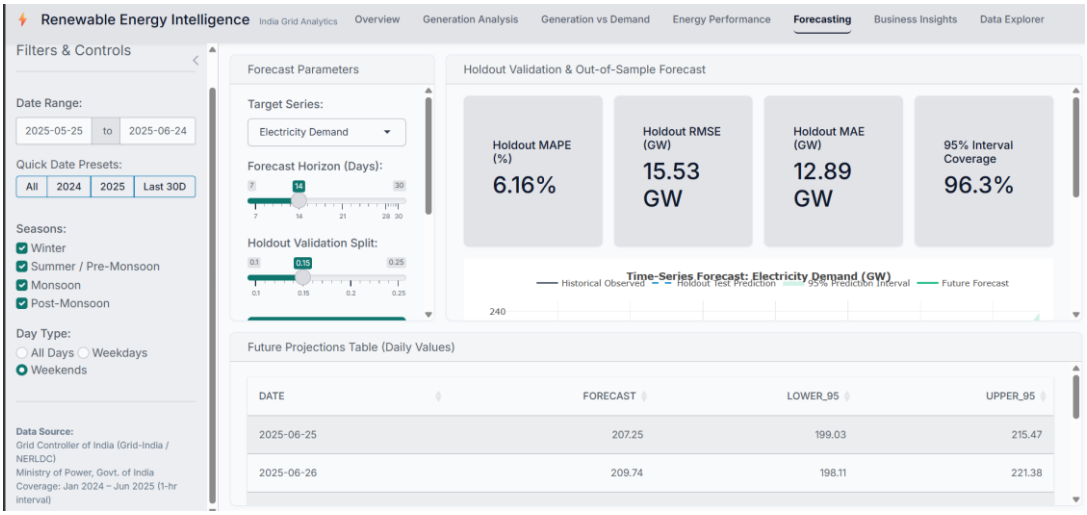
Indicator	Value	Reading
Solar variability (CV)	54.73%	Highest; includes the diurnal cycle if night hours are used
Wind variability (CV)	24.75%	Intermediate
Demand variability (CV)	7.96%	Most stable of the three series
Solar-to-wind correlation (r)	−0.115	Weak negative; descriptive only
Average daytime solar capacity proxy	45.09 GW	Generation-based proxy, not installed capacity

Note: one further indicator row is cut off in the screenshot and is not reported.

7.5 Demand forecast

Figure 6 shows the Forecasting tab and Table 5 interprets its metrics. **Observed:** with a 14-day horizon and a 0.15 holdout split, the daily demand forecast achieved a holdout MAPE of 6.16%, RMSE of 15.53 GW, MAE of 12.89 GW and 96.3% coverage of the 95% prediction interval. The projected demand is 207.25 GW for 25 June 2025 (interval 199.03–215.47 GW) and 207.31 GW for 1 July 2025 (185.43–229.19 GW); the interval width grows from 16.4 to 43.8 GW (derived), and the shaded band in Figure 6(b) widens over the horizon. **Significance:** an average error near 6% is a useful indicative accuracy for planning, and coverage close to the nominal 95% suggests reasonably calibrated intervals, although the number of holdout days is not displayed, so the coverage estimate is itself uncertain. **Limitations:** no benchmark such as a naive forecast is reported, so the 6.16% cannot be judged as good or poor; in the visible part of Figure 6(b) the dashed holdout prediction lies below the observed series, consistent with the MAE but not sufficient to establish bias; the forecast dates extend beyond the dataset end (24 June 2025), so they cannot be validated inside the application; and because the forecast rows include weekdays while Weekends is selected, the Forecasting tab appears not to be restricted by the day-type filter [confirm in the R script].

(a)



(b)

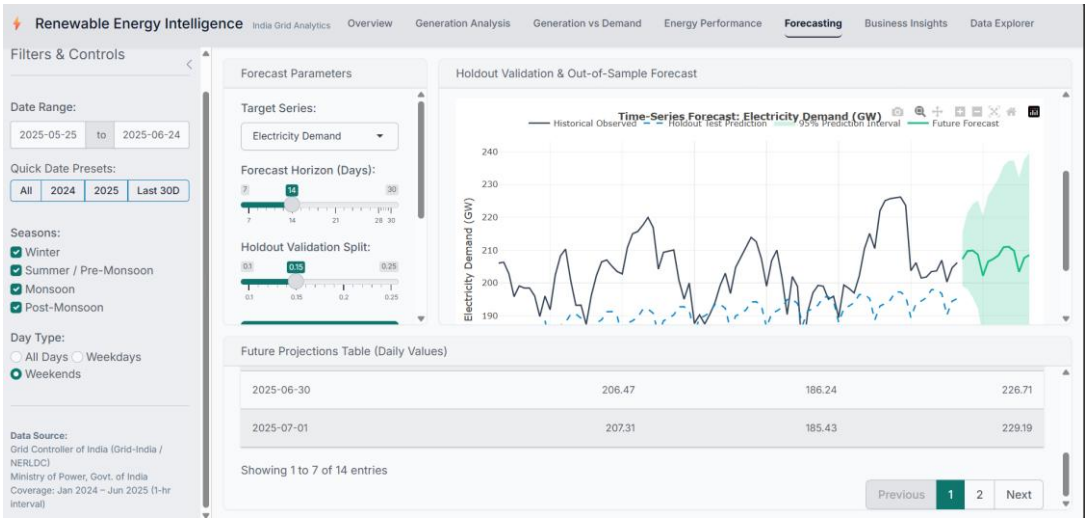


Figure 6. Forecasting tab: (a) forecast parameters, holdout metrics and projection table; (b) holdout validation and out-of-sample forecast chart with 95% prediction interval.

Source: Author's own screenshot of the Renewable Energy Intelligence application, developed using R Shiny.

Table 5. Holdout evaluation metrics for the daily electricity demand forecast

Metric	Value	Reading
Holdout MAPE	6.16%	Average absolute error as a percentage of observed demand
Holdout RMSE	15.53 GW	Error measure that penalises large misses
Holdout MAE	12.89 GW	Average absolute error in GW
95% interval coverage	96.3%	Share of holdout values inside the interval; nominal level 95%

7.6 Business Insights panel

The Business Insights tab (Figure 7) restates the main results as text. It reports a peak solar window at 12:00 IST averaging 46.19 GW and delivering up to 22.6% of instantaneous load, an evening ramp between 17:00 and 20:00 IST averaging +25.92 GW, the solar–wind correlation of −0.115, and a storage window of charging between 10:00 and 14:00 and discharging from 18:30. Two recommendation cards are visible (storage deployment and time-of-day tariffs); further cards are not visible in the screenshot. Four points need reconciliation before submission or reuse. First, the recommendation card quotes solar delivering about 24% of demand whereas the findings card gives 22.6%. Second, the card quotes an evening net ramp of +7 to +38 GW, whereas the maximum one-hour ramp in Figure 2 is +29.11 GW/h; the +38 GW figure may be a multi-hour change but this is not stated. Third, the storage text refers to 35–50 GW of “surplus” solar, yet renewables never exceeded 37% of demand, so no

surplus relative to demand is evidenced. Fourth, the statements about current tariffs are not derived from the dataset. Whether the narrative numbers update with the filters or are fixed text also affects how far they can be trusted.

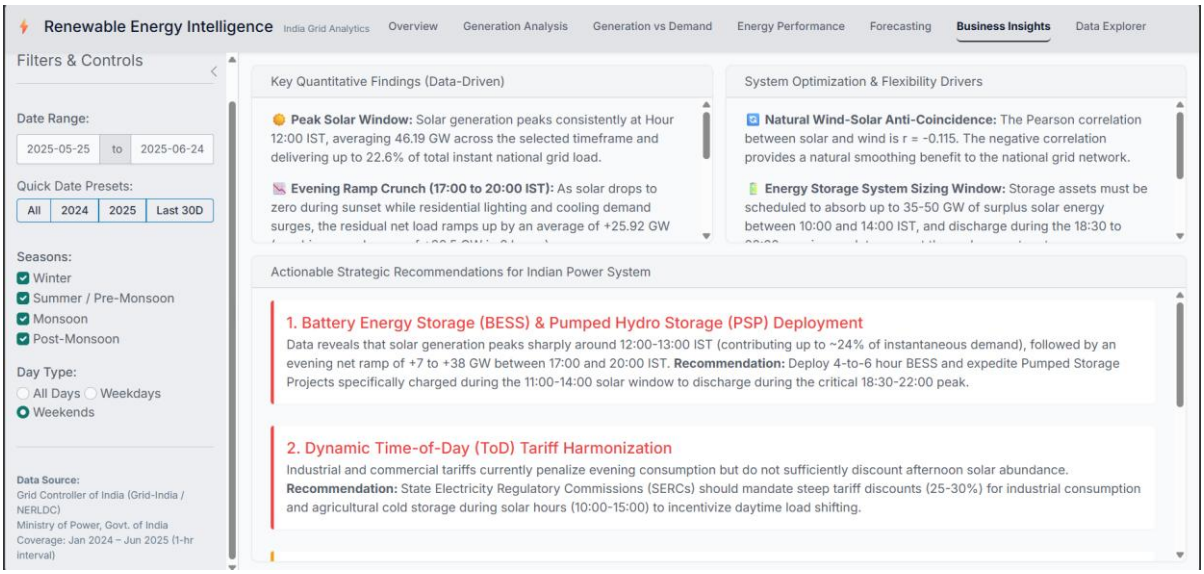


Figure 7. Business Insights tab: key quantitative findings, flexibility drivers and the first strategic recommendations.

Source: Author's own screenshot of the Renewable Energy Intelligence application, developed using R Shiny.

8. Business Insights and Recommendations

The recommendations below follow from the observations in Section 7. Each is labelled by what it still needs before it can be treated as validated; none is presented as a proven finding.

1. **Evaluate storage that shifts midday solar into the evening.** Evidence: solar falls to near zero after sunset, peaks at 61.71 GW, and residual demand rises by up to 29.11 GW in an hour. *Requires validation:* dispatch simulation, curtailment data and storage cost data. The 4–6 hour battery and 35–50 GW figures in the dashboard text are indicative and should not be adopted without modelling; the benefit would come from energy shifting rather than from absorbing demonstrated surplus.
2. **Plan dispatchable ramping capability for the 17:00–20:00 window.** Evidence: the right tail of the ramp distribution (Figure 4) and the average evening ramp of +25.92 GW (Figure 7). *Requires validation:* unit-level ramp limits and sub-hourly data, which this dataset lacks.
3. **Treat wind–solar diversity as limited until tested.** Evidence: r = −0.115 only; wind nevertheless contributed more energy than solar and kept the share above 5% in every hour. *Requires validation:* repeat the correlation by season and day type over the full January 2024–June 2025 data.
4. **Explore load-shifting incentives for flexible daytime demand.** Evidence: solar output is concentrated around midday. *Requires additional data:* flexible-load shares and actual tariff structures from regulatory orders; the dashboard's tariff statements and discount percentages are not derived from the dataset.
5. **Strengthen forecasting before operational use.** Evidence: MAPE of 6.16% with no benchmark and intervals that widen to about ±22 GW by the end of the 14-day horizon. *Requires additional work:* a naive-forecast baseline, rolling-origin evaluation, bias monitoring, weather and calendar variables, and forecasts of solar and wind as well as demand.
6. **Use the peak and low milestone table for monitoring unusual hours.** Evidence: the milestone table (Figure 3) already lists extreme hours. *Requires additional data:* plant-level monitoring and weather data to explain any unusual hour; no cause can be inferred from national hourly data.

9. Limitations and Future Scope

- **Narrow evidence window.** The screenshots cover nine weekend days, two of them in May; conclusions about seasons, weekdays or the full 2024–2025 period are not supported by this report.
- **Aggregation level.** National hourly totals hide regional differences and sub-hourly ramps.

- **Missing variables.** There are no weather, irradiance, installed-capacity, curtailment, storage, plant-level or other-generation data. Consequently, solar losses, conversion efficiency and curtailment cannot be measured, and the generation–demand gap cannot be equated with a supply deficit.
- **Descriptive statistics only.** Correlations and CV values do not establish causal relationships.
- **Forecast uncertainty.** A single holdout, an unreported model and benchmark, and a horizon beyond the observed data limit confidence in the forecast.
- **Documentation basis.** The R source code was not available for this report; features were verified from screenshots only, and some panels (duck curve, timeline, Executive Summary, later insight cards) are partly hidden.

Future work can add weather and irradiance variables, regional or plant-level data, battery dispatch simulation, solar and wind forecasting with exogenous predictors, and a check that the narrative text in Business Insights is computed from the filtered data.

10. Conclusion

The project addresses the need for an interactive view of how solar and wind generation relate to national electricity demand. Using hourly Grid-India data (Mukherjee et al., 2025), the R Shiny dashboard provides KPI, distribution, variability, forecasting and data-export views under shared period, season and day-type filters. In the verified weekend window of 25 May to 24 June 2025, solar and wind supplied 17.65% of demand, wind supplied slightly more energy than solar, solar fell to near zero overnight, and residual demand rose by up to 29.11 GW in one hour, which points to evening ramping and storage as the priority questions. The demand forecast reached a holdout MAPE of 6.16%, although it lacks a benchmark. These are descriptive results from a narrow window; they do not measure shortages or losses. The most useful next step is to re-run the same views on the full dataset, reconcile the Business Insights text with the computed indicators, and add a forecast baseline and weather data.

References

Mukherjee, D., Kalita, K., & Kumar, S. (2025). *Electricity demand, solar and wind generation data (September 2021–June 2025) of India at 1-hour interval* (Version 2) [Data set]. Mendeley Data. https://doi.org/10.17632/y58jknpgs8.2

Posit Software, PBC. (n.d.). *Shiny: Web application framework for R* [Documentation]. Retrieved October 11, 2026, from https://shiny.posit.co/

R Core Team. (n.d.). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. https://www.R-project.org/

Sievert, C., Cheng, J., & Aden-Buie, G. (n.d.). *bslib: Custom 'Bootstrap' 'Sass' themes for 'shiny' and 'rmarkdown'* [R package]. CRAN. Retrieved October 11, 2026, from https://doi.org/10.32614/CRAN.package.bslib

Sievert, C., Parmer, C., Hocking, T., Chamberlain, S., Ram, K., Corvellec, M., & Despouy, P. (n.d.). *plotly: Create interactive web graphics via 'plotly.js'* [R package]. CRAN. Retrieved October 11, 2026, from https://doi.org/10.32614/CRAN.package.plotly

Xie, Y., Cheng, J., Tan, X., & Aden-Buie, G. (n.d.). *DT: A wrapper of the JavaScript library 'DataTables'* [R package]. CRAN. Retrieved October 11, 2026, from https://doi.org/10.32614/CRAN.package.DT